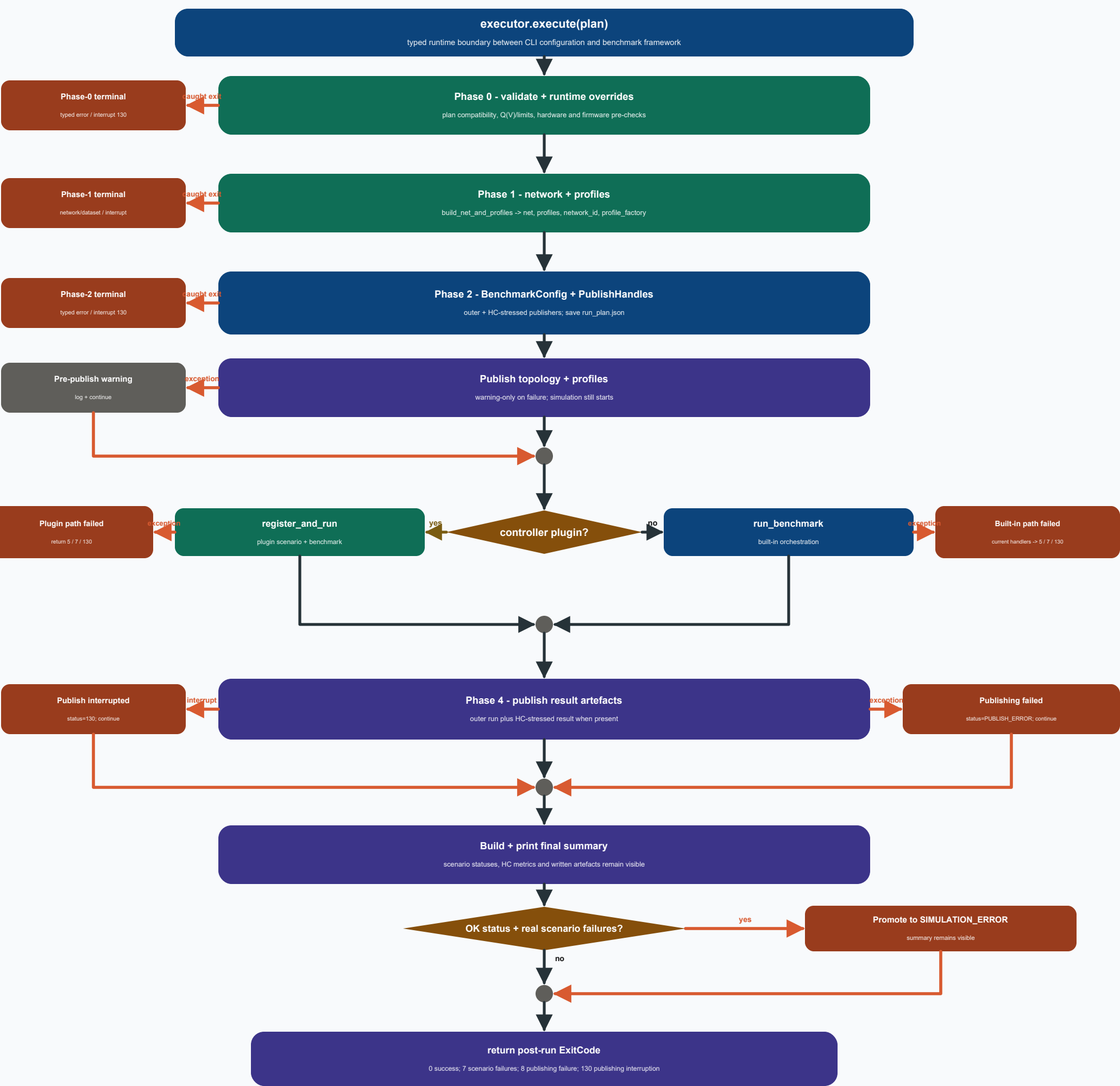




Pre-run safeguards

Validation occurs before a potentially multi-hour run: cross-field constraints, file paths, dataset/network compatibility and serial availability are checked early.

Per-run Q(V) values are applied through `volt_var_controller.set_qv_parameters` so software, coordinator sizing and Arduino CFG use the same characteristic.

Violation-detector thresholds are reset then rebound for the run, keeping default-driven checks consistent with the selected limits.

Hardware is probe-opened only when the chosen scenario set or custom controller will actually use the serial interface.

Run and publication semantics

Network/profile resolution is isolated as its own phase and returns the HC profile factory needed for stressed re-benchmarking.

Topology/profile publication before simulation is intentionally non-fatal, so an export problem does not prevent the benchmark from starting.

A controller plugin routes Phase 3 through `register_and_run`; otherwise the executor invokes `run_benchmark` directly.

A Phase-3 failure or interruption returns immediately; Phase-4 publication problems instead preserve the completed result and continue through the final summary.

Typed process status

CONFIG, NETWORK, DATASET, PLUGIN and HARDWARE failures terminate in their guarded pre-run phases with their typed process status.

Phase-3 controller-definition/import failures map to `PLUGIN_ERROR`; runtime crashes map to `SIMULATION_ERROR`; interruption returns 130 immediately.

Publishing exceptions set `PUBLISH_ERROR` and publishing interruption sets 130, but both still allow the completed benchmark summary to be printed.

If publishing remained OK but `benchmark_runner` reports real scenario failures, the final post-run status is promoted from OK to `SIMULATION_ERROR`.